

SIH 1505 — INDUSTRIAL EMERGENCY PRE-PLAN

INCIDENT ID: INC-T-04-18K  
DOCUMENT VERSION: v1.0 [AUTHORIZED (PROTOTYPE DEMO)]  
FACILITY: PetroChem Complex Alpha - Unit 04  
PROTOTYPE FACILITY REF: PCH-ALPHA-04 (Demo Facility — Non-Statutory Evaluation)  
GENERATED: 2026-08-15 17:27:11 UTC  
PREPARED BY: SIH-1505 Decision Support Engine

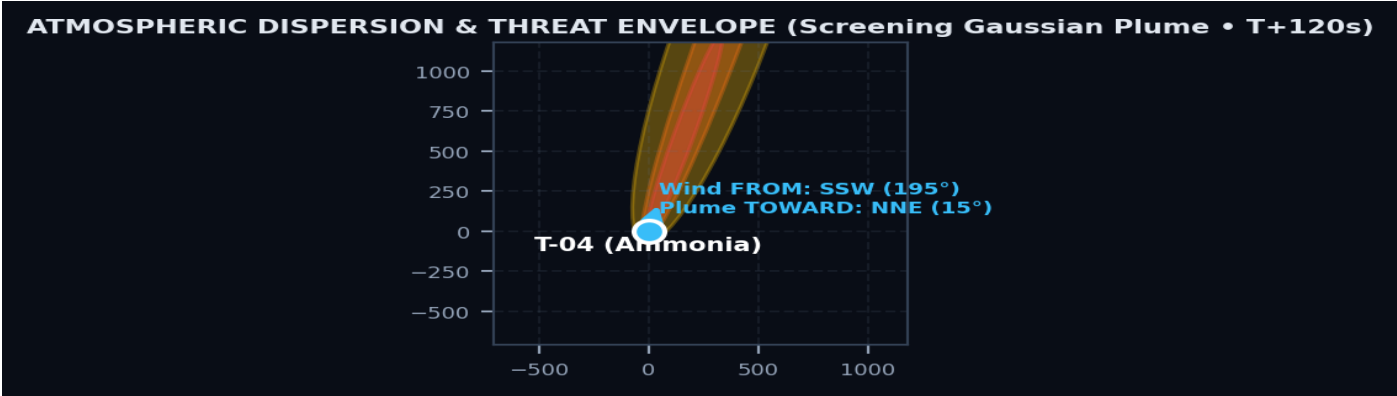
INCIDENT SEVERITY: MODERATE (SCORE: 50.3/100) • GOVERNANCE: AUTHORIZED BY RAMESH THAPA (Incident Commander)

1. INCIDENT SOURCE, CHEMICAL SDS & METEOROLOGICAL PROFILE

|                             |                              |                           |                                             |
|-----------------------------|------------------------------|---------------------------|---------------------------------------------|
| Incident Type:              | Pipeline Leak                | Source Asset:             | T-04 (21.6855, 72.5745)                     |
| Hazardous Substance:        | Ammonia (Anhydrous)          | Release Emission Rate:    | 15.0 kg/s (Duration: 30 min)                |
| Meteorological Wind Vector: | 18.0 km/h • FROM: SSW (195°) | Plume Propagation Vector: | Downwind Travel • TOWARD: NNE (15°)         |
| Ambient Temperature:        | 32.0°C                       | Weather Telemetry Source: | DEMO (Scenario Preset) • Stability: Class D |

2. HAZARD THREAT ZONES (SCREENING GAUSSIAN DISPERSION ENVELOPE)

| Zone Tier          | Threshold Criterion                      | Downwind Reach | Crosswind Width | Threat Area |
|--------------------|------------------------------------------|----------------|-----------------|-------------|
| Red Threat Zone    | ERPG-3 / High Lethality (750.0 ppm)      | 1388.0 m       | 92.2 m          | 83,878 m²   |
| Orange Threat Zone | ERPG-2 / Irreversible Injury (150.0 ppm) | 3000.0 m       | 208.1 m         | 428,999 m²  |
| Yellow Threat Zone | ERPG-1 / Mild Irritation (25.0 ppm)      | 3000.0 m       | 435.0 m         | 732,310 m²  |



3. POPULATION & INFRASTRUCTURE EXPOSURE IMPACT

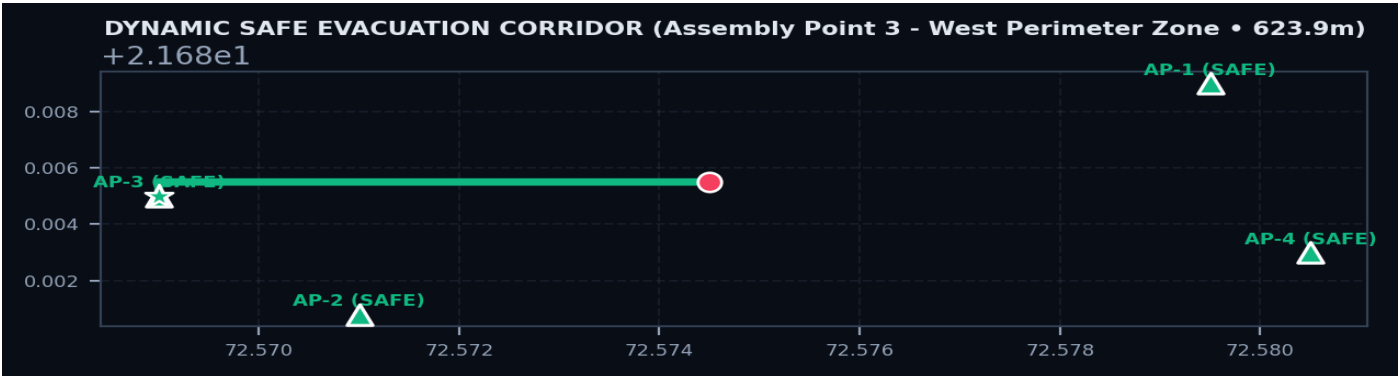
|                               |               |                                |                  |
|-------------------------------|---------------|--------------------------------|------------------|
| Total Workers on Site:        | 28 Personnel  | Lethal Zone (Red) Exposed:     | 1 Workers        |
| Severe Zone (Orange) Exposed: | 0 Workers     | Caution Zone (Yellow) Exposed: | 0 Workers        |
| Compromised Plant Assets:     | 1 of 16 Units | Severed Road Segments:         | 4 of 10 Segments |

4. DYNAMIC SAFE EVACUATION CORRIDOR & MUSTER DIRECTIVES

|                                 |                                                                                                                                                                                                                     |                         |                                                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Designated Safe Assembly Point: | Assembly Point 3 - West Perimeter Zone                                                                                                                                                                              | Exit Perimeter Gate:    | Gate 2 - South Commercial & Tanker Logistics Gate                                                                                       |
| Total Safe Egress Distance:     | 623.9 meters                                                                                                                                                                                                        | Estimated Walk Time:    | 8.7 minutes (at 1.2 m/s)                                                                                                                |
| Multi-Factor Scoring:           | Safety: 70%   Dist: 61%   Rank: 0.664                                                                                                                                                                               | Route Clearance Status: | DIVERTED                                                                                                                                |
| Selection Rationale:            | Selected as optimal Upwind/Crosswind egress corridor (111° clearance from plume axis) providing shortest uncompromised route to Assembly Point 3 - West Perimeter Zone (Safety Score: 0.70, Composite Score: 0.66). | Avoided Blocked Roads:  | Perimeter Ring Road - South, Process Area Cross Road (East-West South), Ammonia & Cryo Tank Service Lane, LPG Sphere Logistics Corridor |

Evaluated Muster Point Alternatives & Rejection Analysis:

| Candidate Assembly Point | Exit Gate | Distance | Safety | Status / Rejection Reason                                                             |
|--------------------------|-----------|----------|--------|---------------------------------------------------------------------------------------|
| ROUTE-AP-3 (AP-3)        | GATE-2    | 623.9 m  | 70%    | SELECTED: None                                                                        |
| ROUTE-AP-2 (AP-2)        | GATE-2    | 884.3 m  | 70%    | VIABLE_BACKUP: Internal access roads severed; requires outer peripheral detour        |
| ROUTE-AP-4 (AP-4)        | GATE-3    | 897.9 m  | 70%    | REJECTED: Internal access roads severed; requires outer peripheral detour             |
| ROUTE-AP-1 (AP-1)        | GATE-1    | 905.8 m  | 25%    | REJECTED: Downwind of toxic plume (38° from plume axis); high vapor inhalation hazard |



5. TACTICAL EMERGENCY RESOURCE DEPLOYMENT & SUPPRESSION STRATEGY

**DECISION-SUPPORT RECOMMENDATION — REQUIRES SITE/HSE VALIDATION:** All tactical quantities, PPE ensembles, suppression demands, and response actions are prototype computational recommendations and must be validated against facility ERDMP and competent safety authorities before operational deployment.

|                          |                                                                                                      |                          |                                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------|
| Upwind Staging Standoff: | 416.4 meters                                                                                         | Isolation Cordon Radius: | 1735.0 meters                                                                                                  |
| Firewater Demand:        | 6,100 LPM                                                                                            | Foam Concentrate Demand: | 0 Liters (AFFF 3%)                                                                                             |
| Mandatory Entry PPE:     | Level A Fully Encapsulated Gas-Tight Suit (Trellechem) with positive-pressure SCBA (NIOSH Certified) | Calculation Basis:       | Calculated from release rate 15.0 kg/s, 1 vulnerable assets, and Ammonia (Anhydrous) SDS toxicity/flammability |

| Unit Name & Type                                                                                                 | Station | Staging Geolocation                                       | Transit | ETA  | Priority  |
|------------------------------------------------------------------------------------------------------------------|---------|-----------------------------------------------------------|---------|------|-----------|
| High-Volume Water Curtain Bowser<br>High-Volume Vapor Cloud Knockdown & Gas Absorption                           | FS-01   | Crosswind Knockdown Sector (105° Vector)                  | 557.1 m | 2.0m | IMMEDIATE |
| Advanced Cardiac & Toxic Resuscitation Ambulance<br>Casualty Triage, Respiratory Resuscitation & Decontamination | MED-01  | Medical Triage at Assembly Point 3 - West Perimeter Zone  | 805.5 m | 2.7m | IMMEDIATE |
| Hazmat Rapid Response Unit (Level A/B Ensembles)<br>Hot Zone Isolation & Emergency Valve Clamping                | FS-01   | Incident Hot-Zone Control Post (15° Upwind)               | 895.2 m | 2.9m | IMMEDIATE |
| Fire Tender Alpha (10,000L Multi-Purpose Water/Foam)<br>Exposure Protection & Boundary Water Curtain Backup      | FS-01   | Upwind Incident Staging Post (Bearing 15°, 416m Standoff) | 895.1 m | 2.9m | HIGH      |

|                                                                                                                                                    |       |                                           |         |      |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------|---------|------|---------|
| <div>Industrial Emergency Response Team<br/>(ERT Alpha)</div> <div>Cordon Enforcement, Road Junction<br/>Barricading &amp; Headcount Control</div> | CR-01 | Main Outer Access Gate & Security Command | 303.6 m | 1.3m | SUPPORT |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------|---------|------|---------|

6. INCIDENT COMMAND EXECUTIVE SUMMARY SNAPSHOT

|                         |                                                         |                     |                                        |
|-------------------------|---------------------------------------------------------|---------------------|----------------------------------------|
| Risk:                   | 50.3/100 — MODERATE                                     | Chemical:           | Ammonia (Anhydrous)                    |
| Personnel Exposed:      | 1 workers (Red: 1)                                      | Compromised Assets: | 1 of 16 units                          |
| Blocked Roads:          | 4 segments                                              | Primary Muster:     | Assembly Point 3 - West Perimeter Zone |
| Exit Gate:              | Gate 2 - South Commercial & Tanker Logistics Gate       | Egress Metrics:     | 623.9 m / 8.7 min                      |
| Lead Tactical Resource: | High-Volume Water Curtain Bowser (IMMEDIATE, ETA: 2.0m) | Firewater Demand:   | 6,100 LPM Water (0 L AFFF Foam)        |

7. STANDARD OPERATING PROCEDURE (SOP) PHASED ACTION DIRECTIVES

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Phase 1: Initial Alarm, Isolation &amp; Personnel Evacuation (0 - 5 Minutes)</b></p> <ul style="list-style-type: none"><li>• Sound Plant Emergency Siren (Continuous Alarm for Ammonia (Anhydrous) Pipeline Leak).</li><li>• Actuate Emergency Shutdown (ESD) for T-04 and isolate upstream pipeline valves.</li><li>• Enforce mandatory crosswind evacuation of all Sector personnel towards Assembly Point 3 - West Perimeter Zone.</li><li>• ERT Team establishes 1735.0m initial cordon perimeter around T-04.</li></ul> <p><b>Phase 2: Source Mitigation &amp; Tactical Exposure Protection (5 - 15 Minutes)</b></p> <ul style="list-style-type: none"><li>• Position Fire Tender &amp; Bowser at Upwind Staging Post (416m standoff, bearing 15°).</li><li>• Establish 6,100 LPM water curtain coverage to suppress Ammonia (Anhydrous) vapors and cool 1 adjacent assets.</li><li>• Deploy Hazmat Entry Squad in Level A Fully Encapsulated Gas-Tight Suit for hot-zone valve isolation and clamp seal.</li><li>• Verify automated fixed water deluge systems on neighboring storage vessels.</li></ul> <p><b>Phase 3: Containment, Medical Triage &amp; Incident De-escalation (15 - 30 Minutes)</b></p> <ul style="list-style-type: none"><li>• Complete headcount verification at Assembly Point 3 - West Perimeter Zone (Reconcile against 1 exposed workers).</li><li>• Ambulance team initiates emergency triage and decontamination for exposed operators.</li><li>• Contain all contaminated firefighting water in industrial effluent retention sump; prevent municipal drain escape.</li><li>• HSE Lead Commander compiles situation briefing for District Disaster Management Authority (DDMA).</li></ul> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

8. DATA TRACEABILITY & SCREENING MODEL ASSUMPTIONS

|                     |                                                                                                      |                      |                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------|
| Dispersion Physics: | Screening Gaussian Approximation with Pasquill-Gifford stability parameters                          | Human Governance:    | Mandatory HSE human review and authorization required prior to operational execution |
| Weather Telemetry:  | DEMO Telemetry (Scenario Preset) • Recorded at 32.0°C, 18.0 km/h                                     | Statutory Scope:     | Decision-support prototype only; non-certified under statutory OISD/ALOHA rules      |
| Terrain Modeling:   | Simplified 2D/3D flat terrain representation with standard surface roughness (z <sub>0</sub> = 0.5m) | Road & Worker Graph: | Prototype facility CAD/GIS network with obstacle avoidance pathfinding               |

9. REGULATORY DISCLAIMER & PROTOTYPE NOTICE

**PROTOTYPE DECISION SUPPORT — NON-CERTIFIED COMPUTATIONAL WORKFLOW:**

This emergency pre-plan document is prepared automatically by the SIH 1505 Decision Support Engine. Dispersion envelopes, routing corridors, and tactical dispatches are computed for hackathon demonstration and decision-support prototyping only. This software does NOT claim certified ALOHA equivalence or statutory compliance. Operational tactical actions require validation against the facility's approved Emergency Response and Disaster Management Plan (ERDMP), applicable OISD/PESO standards, and competent statutory safety authorities.

10. HUMAN-IN-THE-LOOP PRE-PLAN AUTHORIZATION & ENDORSEMENT

**DOCUMENT STATUS: AUTHORIZED (PROTOTYPE DEMO)**

Document Version: v1.0

Prepared by: SIH-1505 Decision Support Engine

Demonstration Approver: Ramesh Thapa

Role / Designation: Incident Commander

Authorization Record ID: AUTH-20260815-0EA43C

Authorized At: 2026-08-15 17:27:11 UTC

Approval Notes: Primary Demo Scenario Authorized.

**DEMONSTRATION SIGNATURE BLOCK:**

✓ **AUTHORIZED (PROTOTYPE DEMO)**

Ramesh Thapa

Incident Commander

DEMO SIGNATURE — NOT A REAL SIGNATURE

(Ready for PKI Digital Signature Integration)